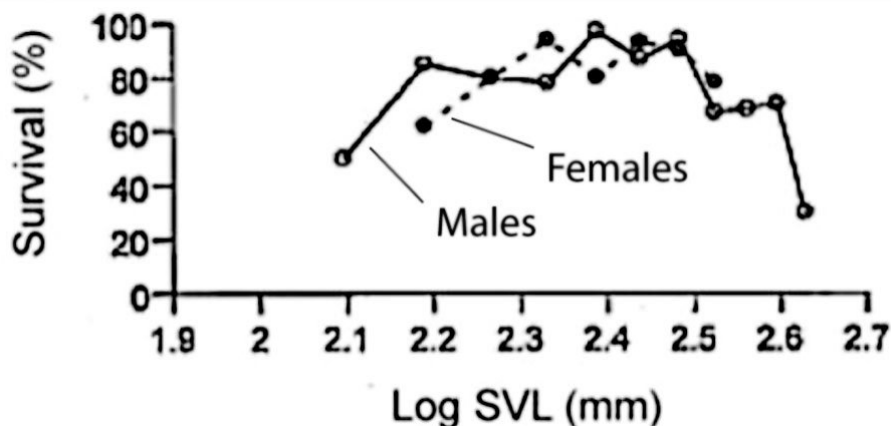


MULTIPLE CHOICE. Choose the one alternative that best completes the statement or answers the question.

- 1) Cystic fibrosis is a genetic disorder in humans where the person has two homozygous recessive alleles for the gene. If the disease is left untreated, it causes severe health problems in the individual. If 9 in 10,000 newborn babies have the disease, what are the expected frequencies of the dominant (A^1) and recessive (A^2) alleles according to the Hardy-Weinberg equation?
A) $f(A^1) = 0.9997$, $f(A^2) = 0.0003$ B) $f(A^1) = 0.9700$, $f(A^2) = 0.0300$
C) $f(A^1) = 0.9604$, $f(A^2) = 0.0392$ D) $f(A^1) = 0.9800$, $f(A^2) = 0.0200$
- 2) Natural selection changes allele frequencies because some _____ survive and reproduce better than others.
A) loci B) individuals C) species D) alleles
- 3) One population of a species has 25 individuals, all with genotype AA; a second population of this species has 40 individuals, all with genotype aa. Assume that these populations live far apart but in similar environmental conditions. Based on this information, the observed genetic variation most likely resulted from _____.
A) gene flow B) directional selection
C) genetic drift D) nonrandom mating
- 4) No two people are genetically identical, except for identical twins. The main source of genetic variation among humans is _____.
A) environmental effects
B) the reshuffling of alleles in sexual reproduction
C) genetic drift
D) new mutations that occurred in the preceding generation
- 5) If the nucleotide variability of a locus equals 0%, what is the gene variability and number of alleles at that locus?
A) gene variability > 0%; number of alleles = 2 B) gene variability = 0%; number of alleles = 1
C) gene variability = 0%; number of alleles = 0 D) gene variability = 0%; number of alleles = 2
- 6) A fruit fly population has a gene with two alleles, A^1 and A^2 . Tests show that 70% of the gametes produced in the population contain the A^1 allele. If the population is in Hardy-Weinberg equilibrium, what proportion of the flies carry both A^1 and A^2 ?
A) 0.21 B) 0.49 C) 0.42 D) 0.7
- 7) The value of the Hardy-Weinberg equation is that it allows scientists to perform which of the following assessments?
A) to provide proof that natural selection is acting
B) to measure the effects of gene flow among populations
C) to detect whether an evolutionary force is acting on a population
D) to predict the percentage of people in a population that is homozygous for the recessive allele that causes a genetic disease
- 8) In a Hardy-Weinberg population with two alleles A^1 and A^2 that are in equilibrium, the frequency of allele A^2 is 0.1. What is the frequency of individuals with A^1A^1 genotype?
A) 0.42 B) 0.20 C) 0.81 D) 0.32

- 9) Two scientists measured the snout-to-vent (anus) length of Galápagos marine iguanas (lizards) and observed the percent survival of different-sized animals, all of the same age. The graph shows the log snout-vent length (SVL, a measure of overall body size) plotted against the percent survival of these different size classes for males and females.



(Data from M. Wikelski and L. Michael Romero. Body size, performance and fitness in Galapagos marine iguanas. *Integrative and Comparative Biology* 43:376-386 [2003].)

Currently, the only predators of Galápagos marine iguanas are Galápagos hawks. Although small iguanas can sprint faster than large iguanas, iguana body size is not correlated with risk of hawk predation. If predators (for example, cats) that preferentially catch and eat slower iguanas are introduced to the island, iguana body size is likely to _____ in the absence of other factors; the iguanas would then be under _____ selection.

- A) decrease; directional
B) increase; directional
C) decrease; disruptive
D) increase; disruptive
- 10) Imagine a human population with an extremely low frequency of sickle cell alleles. Which of the following changes is most likely in the human population 15 generations after introducing a mosquito population that carries the malaria parasite?
- A) a decrease in the number of mosquito bites
B) an increase in red blood cells without nuclei
C) an increase in the mutation rate of the hemoglobin gene
D) an increase in the sickle cell allele
- 11) A biologist doing a long-term study on a wild spider population observes increased variation in silk thickness. Which of the following processes is the most likely to be affecting the spider population?
- A) directional selection
B) disruptive selection
C) stabilizing selection
D) genetic drift
- 12) In which of the following situations would genetic drift most likely cause evolution?
- A) The population has inherited traits better suited to a different environment.
B) A sudden change in the environment drastically increases the population size.
C) Chance events cause allele frequencies to fluctuate unpredictably.
D) The size of the gene pool increases because a smaller group establishes a new population.
- 13) Most Swiss starlings (birds) produce four to five eggs in each clutch. Starlings producing more or less eggs have reduced fitness. Which of the following terms best describes this situation?
- A) stabilizing selection
B) disruptive selection
C) sexual selection
D) directional selection

- 14) A proficient engineer can easily design skeletal structures that are more functional than those currently found in the forelimbs of such diverse mammals as horses, whales, and bats. Which of the following statements best explains why the actual forelimbs of these mammals do not seem to be optimally arranged?
- A) Natural selection is generally limited to modifying structures that were present in previous generations and in previous species.
 - B) Natural selection has not had sufficient time to create the optimal design in each case, but will do so given enough time.
 - C) Although we may not consider the fit between the current skeletal arrangements and their functions excellent, we should not doubt that natural selection ultimately produces the best design.
 - D) In many cases, phenotype is determined by genotype and the environment.
- 15) In those parts of equatorial Africa where the malaria parasite is most common, the sickle-cell allele constitutes 20% of the β hemoglobin alleles in the human gene pool.

In the United States, the parasite that causes malaria is not present, but the sickle cell allele is present in many African-Americans whose ancestors were from equatorial Africa. Which of the following processes are most likely acting on the sickle-cell allele in the United States, and in equatorial Africa?

- A) stabilizing selection in the United States; disruptive selection in equatorial Africa
 - B) directional selection in the United States; stabilizing selection in equatorial Africa
 - C) disruptive selection in the United States; stabilizing selection in equatorial Africa
 - D) directional selection in the United States; disruptive selection in equatorial Africa
- 16) Suppose 64% of the individuals in a remote mountain village can taste phenylthiocarbamide (PTC) and must, therefore, have at least one copy of the dominant PTC taster allele. If this population conforms to Hardy-Weinberg equilibrium for this gene, what percent of the population must be heterozygous for this trait?
- A) 32%
 - B) 40%
 - C) 16%
 - D) 48%
- 17) Researchers studying a small milkweed population note that some plants produce a toxin and other plants do not. They identify the gene responsible for toxin production. One allele (T^1) codes for an enzyme that makes the toxin, and another allele (T^2) codes for a nonfunctional enzyme that cannot produce the toxin. Heterozygotes produce an intermediate amount of toxin. The researchers measured the abundance of each of the three possible genotypes and compared those numbers to the expected numbers if the population were in Hardy-Weinberg equilibrium. Those numbers are shown in the chart.

	T^1T^1	T^1T^2	T^2T^2
Hardy-Weinberg expected numbers	98	84	18
Observed numbers	112	56	32

Based on these data, which answer correctly identifies whether the population is likely in Hardy-Weinberg equilibrium?

- A) No, there are more heterozygotes than expected.
 - B) Yes, the population appears to be in Hardy-Weinberg equilibrium.
 - C) No, there are more homozygotes than expected.
 - D) More information is needed to answer this question.
- 18) In a Hardy-Weinberg population with two alleles A^1 and A^2 that are in equilibrium, the frequency of the allele A^2 is 0.3. What is the frequency of individuals that are homozygous for this allele?
- A) 0.9
 - B) 9.0
 - C) 0.49
 - D) 0.09

- 19) Restriction enzymes in bacteria protect the bacteria from successful attack by bacteriophages, whose genomes can be degraded by the restriction enzymes. The bacterial genomes are not vulnerable to these restriction enzymes because bacterial DNA is methylated. This situation selects for bacteriophages whose genomes are also methylated. As new strains of resistant bacteriophages become more prevalent, they in turn select for bacteria whose genomes are not methylated and whose restriction enzymes instead degrade methylated DNA. Which of the following factors is most likely to cause the changes?

A) evolutionary imbalance
B) heterozygote advantage
C) neutral variation
D) frequency-dependent selection

- 20) A species of moth lays eggs on oak trees in spring and summer. Spring caterpillars eat oak flowers and look like oak flowers. Summer caterpillars eat oak leaves and look like twigs. To investigate how the same population of moths can produce such variable caterpillars, a researcher collected eggs from a population of moths and allowed them to hatch and grow under the conditions shown in the table below.

Treatment	Temperature	Day Length	Food
1	Spring-like	Spring-like	Oak flowers
2	Spring-like	Spring-like	Oak leaves
3	Spring-like	Summer-like	Oak flowers
4	Spring-like	Summer-like	Oak leaves
5	Summer-like	Spring-like	Oak flowers
6	Summer-like	Spring-like	Oak leaves
7	Summer-like	Summer-like	Oak flowers
8	Summer-like	Summer-like	Oak leaves

Comparison of which treatments would allow the researcher to assess the effect of food type on caterpillar morphology?

A) 1, 2, 4, and 5 versus 3, 6, 7, and 8
B) 1, 2, 3, and 4 versus 5, 6, 7, and 8
C) 1, 3, 7, and 8 versus 2, 4, 6, and 8
D) 1, 3, 5, and 7 versus 2, 4, 6, and 8

Answer Key

Testname: UNTITLED1

- 1) B
ID: bio12-- 23.1-10
- 2) B
ID: bio12-- 23.2-1
- 3) C
ID: bio12-- 23.2-4
- 4) B
ID: bio12-- 23.2-2
- 5) B
ID: bio12-- 23.2-3
- 6) C
ID: bio12-- 23.2-5
- 7) C
ID: bio12-- 23.1-21
- 8) C
ID: bio12-- 23.1-24
- 9) A
ID: bio12-- 23.1-45
- 10) D
ID: bio12-- 23.1-43
- 11) B
ID: bio12-- 23.1-47
- 12) C
ID: bio12-- 23.1-30
- 13) A
ID: bio12-- 23.1-52
- 14) A
ID: bio12-- 23.1-54
- 15) B
ID: bio12-- 23.1-56
- 16) D
ID: bio12-- 23.1-11
- 17) C
ID: bio12-- 23.1-16
- 18) D
ID: bio12-- 23.1-22
- 19) D
ID: bio12-- 23.1-49
- 20) D
ID: bio12-- 23.1-6